Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Neurosymbolic Framework for Explainable Alternative Credit Decisioning			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (✓), AI (✓), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Artificial Intelligence		Cybersecurity	
	Medium Category	Small Category	Medium Category	Small Category
	Advanced AI modeling/decision-making	Safe and reliable AI	Data / AI security	Digital vulnerability analysis/response
Concept of Proposed Technology	This project proposes a neurosymbolic credit decisioning framework that integrates large language models (LLMs), knowledge graphs (KGs), and causal graphs (CGs) to enable trustworthy, explainable, and privacy-preserving alternative credit assessment. The system leverages non-traditional data sources, such as SMS-derived entities, transaction metadata, and text messages, while strictly avoiding direct access to raw personal data. A key innovation of the proposed framework is that all unstructured alternative data are transformed into abstracted semantic and causal representations prior to any learning or inference. LLMs are used only to extract high-level entities, relations, and behavioral signals, which are then encoded			

	into structured KGs and CGs. Unlike conventional black-box credit models, the proposed system performs decisioning through explicit symbolic reasoning over semantic and causal structures, producing verifiable reasoning traces that can be independently inspected and validated. These traces make it possible to explain why a decision was made, which abstracted factors contributed, and how causal assumptions influenced outcomes, enabling transparency that is suitable for internal risk teams, consumer-facing explanations, and regulatory audits. By operating exclusively on masked, abstracted features and grounding decisions in explicit causal and domain knowledge, the framework aligns with strict data protection and fair-lending requirements while enabling personalized credit assessments. The result is a scalable alternative credit decisioning system that reconciles predictive performance, explainability, and regulatory compliance: three properties that are typically in tension in existing credit scoring approaches.
Rationale for Joint R&D Project	Financial technology SMEs face increasing pressure to deploy high-performing AI-driven credit decisioning systems that are simultaneously explainable, auditable, and compliant with rapidly evolving regulations governing automated decision-making. Meeting these requirements has become particularly challenging as regulators demand not only accuracy, but also transparency, traceability, and demonstrable fairness in credit decisions, capabilities that exceed the scope of conventional machine-learning pipelines. This joint R&D project leverages the complementary strengths of the partnering Korean SME(s) and RPI. The Korean SME partner is expected to contribute domain expertise in real-world credit operations, and proven market deployment capabilities, ensuring that the proposed technology is grounded in practical constraints and commercialization pathways. In contrast, RPI brings foundational research expertise in causal reasoning, KG–based explainability, and privacy-preserving AI architectures, including neurosymbolic methods that are difficult for SMEs to develop internally due to their technical depth and long development horizons. By combining applied industrial capabilities with advanced academic research, this collaboration enables the co-development of regulator-aligned, export-ready alternative credit decisioning technology that neither partner could achieve independently. The joint framework accelerates technology transfer from research to practice, reduces regulatory and technical risk for market deployment, and positions the resulting system for international adoption across jurisdictions with stringent financial and data protection regulations.
Key Words	1. Explainable Artificial Intelligence (XAI) 2. Causal Reasoning 3. Knowledge Graphs 4. Credit Risk Assessment 5. Responsible AI in Finance

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (2)	**End of the Project**	Level (6)
R&D Objectives	1. Design a privacy-preserving knowledge graph representation for alternative credit data. 2. Develop causal credit graphs enabling counterfactual and causal explanations. 3. Integrate constrained LLMs for regulator-grade explanation generation. 4. Validate fairness, robustness, and auditability of credit decisions.			
R&D Target Specifications	1. Zero raw data exposure will be enforced at the system, storage, and model interface levels. 2. 100% decision traceability via KG reasoning paths, where every decision produces a machine-verifiable reasoning path. 3. Improved explanation consistency compared to state-of-the-art baselines, measured via explanation stability, fidelity, and regulator interpretability studies. 4. Demonstrated regulatory audit readiness via mock audits and documentation artifacts.			
R&D Plan	This joint R&D project will be executed in two phases. Phase I focuses on foundational design, feasibility validation, and risk reduction. Phase II focuses on scalable development, integration, pilot deployment, and commercialization readiness. The overall approach emphasizes privacy-preserving AI, causal reasoning, and explainability using only abstracted and masked data. **Architectural Overview and Design Rationale.** As illustrated in Fig. 1, the system is explicitly partitioned into a Trusted Zone and an Inference Zone, enforcing privacy-by-design separation between raw alternative data and downstream decisioning. All unstructured data processing, including entity extraction, abstraction, and personal knowledge graph construction, occurs locally within the Trusted Zone. Only abstracted, provenance-annotated symbolic representations cross into the Inference Zone, where causal reasoning, credit scoring, and explanation generation occur. This architectural separation operationalizes data minimization principles while enabling verifiable reasoning and scalable deployment across institutional boundaries.			

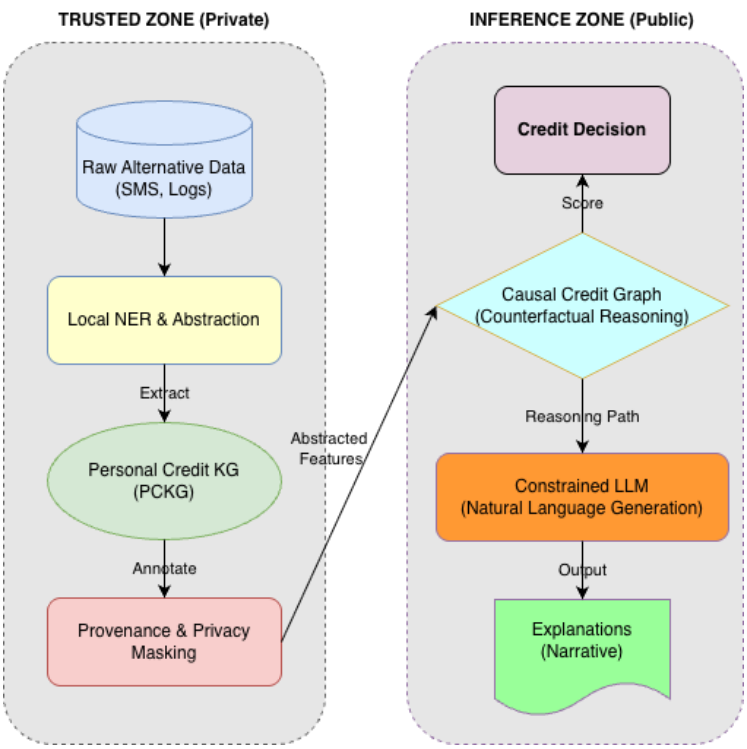

Fig 1: Proposed System Architecture

Phase I Methodology (6 Months): Foundational Models, Formalization, and Feasibility

Phase I emphasizes formal representation design, causal model specification, and explanation fidelity validation, with the explicit goal of reducing downstream regulatory and deployment risk. The focus is not on maximizing predictive performance, but on establishing semantically grounded, causally meaningful, and auditable decision structures that can scale safely in Phase II.

1. Semantic Abstraction and Knowledge Graph Modeling

We adopt a KG–first representation of alternative credit data, where abstracted NER outputs (e.g., EMI amount, due date, payment status) are mapped to formally defined semantic entities and relations. A KG-first representation enables explicit modeling of temporal, relational, and provenance semantics that are not expressible in flat feature vectors, forming the foundation for causal reasoning and explanation traceability.

Methodologically, this follows established approaches in:

- Explanation Ontology (EO) for structuring explainable concepts and reasoning artifacts, enabling machine-interpretable explanations rather than free-form narratives (Chari et al., 2024).
- Personal Knowledge Graph (PKG) construction, previously applied by the PI to health and finance domains, where user-centric, temporally grounded graphs enable individualized reasoning without raw data exposure (Seneviratne & Shukla, 2023 and Spadea & Seneviratne 2025).

The resulting Personal Credit Knowledge Graph (PCKG) encodes:

- financial entities (loans, EMIs, lenders),
- behavioral events (payments, overdue events),
- temporal relations (recency, frequency),
- and provenance metadata describing extraction confidence and masking level.

This KG-first abstraction strategy builds directly on the PI's prior work on knowledge-enabled

explainable systems and user-centric graph representations, where semantic structure enables faithful reasoning and explanation beyond feature attribution methods (Chari et al., 2020; Chari et al., 2024). Unlike embedding-only approaches, the PCKG preserves explicit temporal, relational, and provenance semantics, which are essential for causal modeling, explanation, and longitudinal credit assessment.

2. Provenance-Aware Feature Governance

All KG entities and relations are annotated with provenance metadata, following the PI's prior work on accountable systems and provenance in data science (Seneviratne, 2021; Sikos et al., 2021). Provenance metadata also enables selective exclusion or reweighting of low-confidence features, reducing model brittleness and regulatory risk. Provenance annotations further enable decision-time accountability, allowing the system to justify not only which factors influenced an outcome, but also how reliable, how recent, and under what extraction constraints those factors were derived.

Methodologically, provenance is used to:

- trace each decision outcome back to contributing abstracted features,
- support internal audits without raw data access,
- enable regulator-facing transparency on "what information was used and why."

3. Causal Credit Graph Construction

To move beyond correlation-based explanations, we construct a Causal Credit Graph (CCG) layered on top of the PCKG to ensure fairness. For example, a causal graph would allow us to detect and prune biased paths (e.g., checking if Zip Code acts as a proxy for Race). The CCG explicitly distinguishes between immutable attributes, behavioral variables, and actionable intervention variables, ensuring compliance with fair lending and explainability requirements.

Methodologically, causal modeling proceeds as follows:

- Identify causally meaningful variables (e.g., repeated missed EMIs, payment regularity, lender diversity).
- Encode causal assumptions using directed acyclic graphs (DAGs) grounded in domain knowledge.
- Validate causal structures through sensitivity analysis and counterfactual testing.

This approach aligns with prior knowledge-enabled decision support systems developed by the PI, where causal relations are explicitly represented and queried rather than implicitly learned (Chari et al., 2023; Seneviratne et al., 2023). Importantly, the explicit causal structure enables fairness-aware pruning and constraint enforcement, ensuring that protected attributes and their proxies cannot influence credit decisions except through legally permissible pathways. This design directly addresses regulatory concerns regarding indirect discrimination and spurious correlations, which are difficult to detect in purely statistical models.

4. Structured Explainability via Knowledge-Enabled Reasoning

Instead of post-hoc feature attribution (e.g., SHAP), explanations are generated as explicit reasoning paths through the KG and CCG:

Entity → Relation → Temporal Context → Causal Effect → Decision Outcome

This methodology is grounded in the PI's prior work on knowledge-enabled explainable AI, where explanations are structured, reproducible, and semantically verifiable (Chari et al., 2020; Chari et al., 2024). Large Language Models (LLMs) are used only as a constrained interface, translating verified reasoning paths into human-readable explanations, where the LLM is strictly constrained.

By construction, the LLM has no access to raw data, latent embeddings, or scoring logic, and cannot introduce new facts or reasoning steps, its output space is strictly limited to narrativizing pre-validated symbolic reasoning paths as demonstrated in PI's prior work (Mridul & Seneviratne, 2025).

Phase II Methodology (2.5 Years): Scaling, Integration, and Validation

Phase II focuses on scaling, cross-institutional learning, and real-world validation, with explicit attention to deployment constraints faced by financial SMEs, including data residency laws, model governance requirements, and operational monitoring.

5. Scalable Knowledge Graph Engineering

Phase II extends the PCKG to production scale using modular schema design and incremental graph updates. This methodology draws on PI's prior work in sustainable and evolving knowledge graphs (Spadea & Seneviratne, 2025), ensuring: schema extensibility across markets and lenders, robustness to concept drift, and efficient querying for real-time decisioning. Graph quality is continuously assessed using metrics for completeness, consistency, and temporal coherence, in accordance with best practices established in the PI's KG research. Schema modularity enables rapid adaptation to new regulatory regimes and market contexts without retraining core models.

6. Advanced Causal and Counterfactual Reasoning

These counterfactual explanations support both internal risk analysis and consumer-facing adverse action explanations.

Causal reasoning is extended to support:

- subgroup-specific causal models,
- temporal counterfactual queries (e.g., "What if the overdue event had occurred three months earlier?"),
- policy-aligned what-if analysis.

This builds on methodologies previously applied by the PI in risk analysis and decision-support systems, where causal explanations improve trust and decision robustness beyond statistical correlations (Green et al., 2023; Seneviratne et al., 2023).

7. Privacy-Preserving and Federated Learning Integration

To enable learning across institutions and markets without data sharing, we integrate federated and vertically partitioned learning techniques operating solely on abstracted features. We implement a Hub-and-Spoke Federated Architecture. The RPI-designed global model aggregates logical rules and causal weights, while local nodes compute gradients on private SME data. This allows the model to learn 'What makes a good borrower' generally, without seeing specific borrower data. Federated aggregation operates over causal parameters and symbolic constraints, rather than opaque model weights, enabling meaningful global learning while preserving local autonomy and regulatory interpretability (Carta et al., 2025).

Methodologically, this follows prior work by the PI on:

- privacy-preserving federated learning for financial crime detection (Khan et al., 2024),
- data distillation and representation learning without raw data exchange (Kang et al., 2025).

This ensures compliance with data residency laws, protection of SME intellectual property, and scalable model improvement without centralized data aggregation. Operating exclusively on abstracted representations significantly reduces vulnerability to membership inference and reconstruction attacks.

	8. Multi-stage Offline Validation and Online Testing The proposed system undergoes a rigorous evaluation lifecycle that transitions from laboratory-scale prototyping to real-world production deployment. Methodologically, this begins with Offline Evaluation for rapid iteration, utilizing small-scale "playground" datasets and large-scale simulated environments to ensure accurate performance benchmarking and robustness. Upon establishing a production-ready API, we implement Shadow Testing to verify consistency between offline simulations and live data. Final validation is conducted through Online Evaluation (A/B Testing) via a canary deployment strategy, starting with a 5% routing portion. This phased approach allows for the integration of real-world feedback and gradual scaling based on online performance metrics, ensuring the system's reliability and stability in complex financial environments. In addition to standard predictive metrics, evaluation will include: - explanation stability under data perturbation, - causal consistency across subgroups, - provenance completeness for audit trails, - and alignment between automated explanations and human expert assessments. Collectively, this phased R&D plan advances a neurosymbolic alternative credit decisioning paradigm in which privacy, explainability, and causal validity are first-class design constraints rather than post-hoc add-ons. By grounding learning and explanation in structured knowledge, causal reasoning, and provenance-aware governance, the proposed system establishes a technically rigorous and regulator-aligned foundation for scalable, trustworthy credit decisioning.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	RPI will lead the foundational research, methodological design, and scientific validation of the proposed neurosymbolic credit decisioning framework, which includes the following R&Rs. 1. Core Architecture and Methodological Design: • Design the overall neurosymbolic system architecture, including the separation of trusted and inference zones. • Define the KG and CG schemas, abstraction pipelines, and explanation interfaces. 2. Causal Modeling and Explainability Frameworks • Develop CCGs enabling counterfactual and policy-aligned reasoning. • Design structured, regulator-verifiable explanation mechanisms grounded in KGs and the EO. 3. Privacy-Preserving and Federated Learning Methods • Lead the design of privacy-preserving learning mechanisms, including federated and vertically partitioned learning over abstracted features. • Develop safeguards against inference, reconstruction, and membership attacks. 4. Formal Evaluation and Validation Methodology • Define evaluation protocols for fairness, robustness, explainability fidelity, and auditability. • Conduct controlled experiments and sensitivity analyses to validate causal assumptions and explanation stability. 5. Knowledge Transfer and Technical Guidance • Provide technical documentation, reference implementations, and architectural guidelines to support SME adoption. • Mentor SME engineers through joint design reviews and code walkthroughs to ensure faithful technology transfer.

Expected R&R of Korean SME Partner	The Korean SME partner will lead the applied development, integration, and commercialization-oriented validation of the proposed system, ensuring alignment with real-world credit operations, regulatory constraints, and market requirements, which includes the following R&Rs: 1. Industrial Data Pipelines and Domain Expertise • Provide access to representative alternative credit data pipelines (e.g., SMS-derived features, transaction metadata) within secure, compliant environments. • Contribute domain knowledge on lending workflows, credit policies, and operational constraints. 2. System Integration and Engineering • Implement production-grade data ingestion, abstraction, and deployment pipelines consistent with the proposed architecture. • Integrate the neurosymbolic decisioning components with existing credit scoring and risk management systems. 3. Pilot Deployment and Real-World Validation • Conduct pilot deployments in controlled production settings, including shadow testing and staged rollouts. • Collect operational performance metrics and feedback from risk teams, compliance officers, and end users. 4. Regulatory and Market Alignment • Ensure alignment with domestic and international regulatory requirements relevant to automated credit decisioning. • Provide feedback on explanation usability, compliance reporting needs, and operational feasibility. 5. Commercialization and Export Readiness • Lead productization efforts, including API hardening, monitoring, documentation, and compliance reporting tools. • Develop commercialization strategies targeting international markets, leveraging the system's explainability and privacy-preserving design as differentiators.

5. Expected Outcome/Impact

Expected Outcomes and Impact	**Expected Outcomes** • Prototype neurosymbolic credit decisioning system integrating KGs, CGs, and constrained LLM-based explanation generation over abstracted/masked alternative data. • PCKG + CCG designs (schemas, reasoning rules, provenance annotations) enabling counterfactual explanations and audit-ready reasoning traces. • Software for privacy-preserving learning and governance components, including federated/partitioned learning over abstracted features plus monitoring artifacts for fairness, robustness, and auditability. • Deployment-ready toolkit (APIs, documentation, reference implementation) transferred jointly to support SME integration and pilot deployment. • High quality publications **Expected Impact** • Trustworthy and regulator-aligned AI decisioning: verifiable explanations and provenance-backed audit trails reduce compliance risk and increase transparency. • Commercial and export potential for the SME: faster productization of differentiated, compliant alternative credit capabilities. • Broader scientific and societal benefit: advances neurosymbolic XAI for high-stakes finance and supports fairer access to credit for thin-file/underbanked consumers

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization